

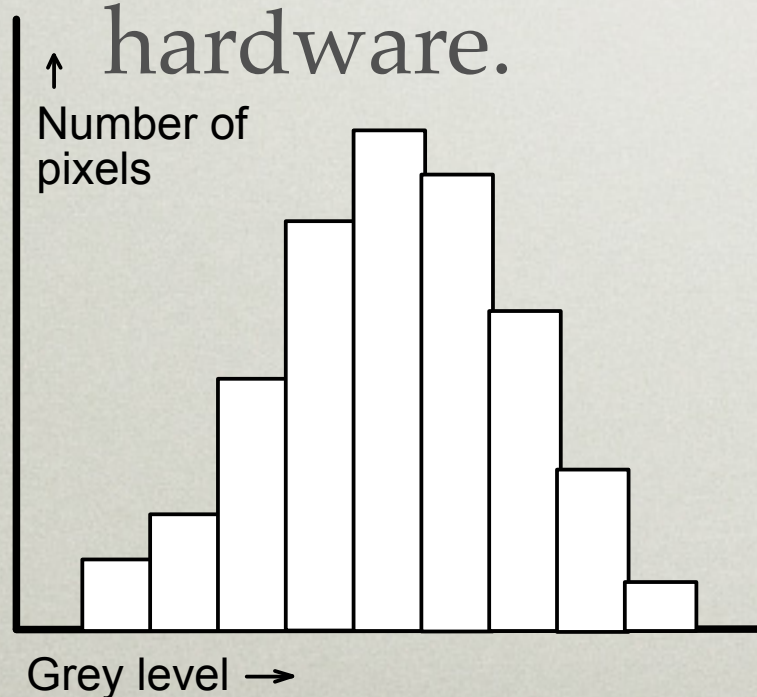
Introduction to Image Processing

BASIC OPERATIONS

- The grey level histogram
- Grey level histogram equalization
- Point operations
- Algebraic operations

THE GREY-LEVEL HISTOGRAM

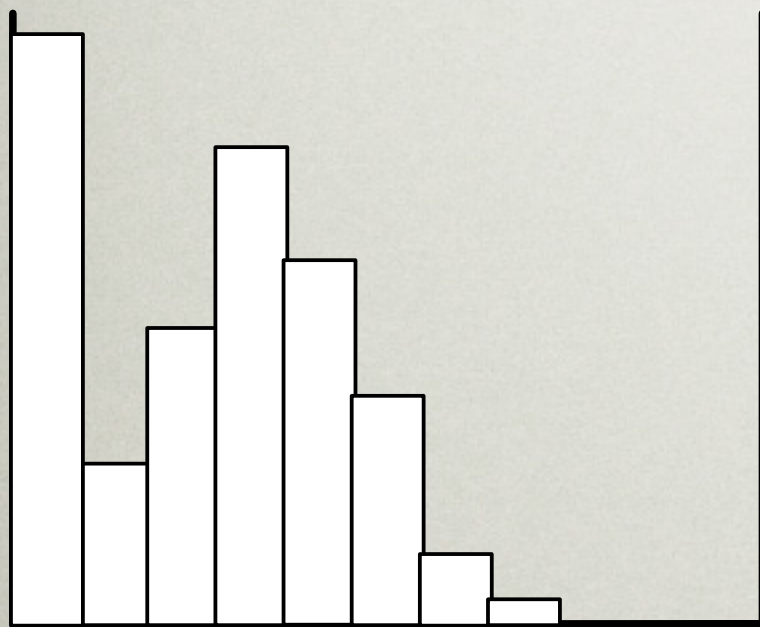
- One of the simplest, yet most useful tools.
- Can show up faulty settings in an image digitizer.
- Almost impossible to achieve without digital hardware.



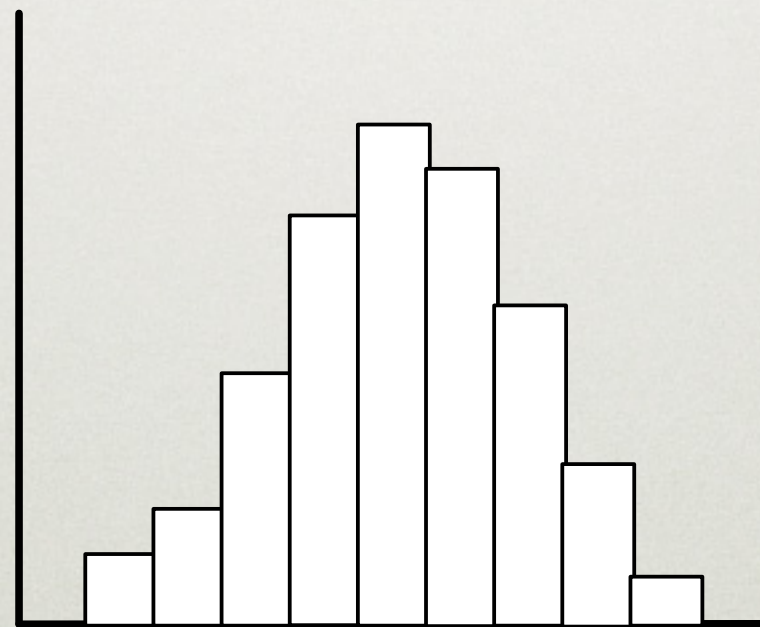
The GLH is a function showing for each grey level the number of pixels that have that grey level.

CORRECTING DIGITISER SETTINGS

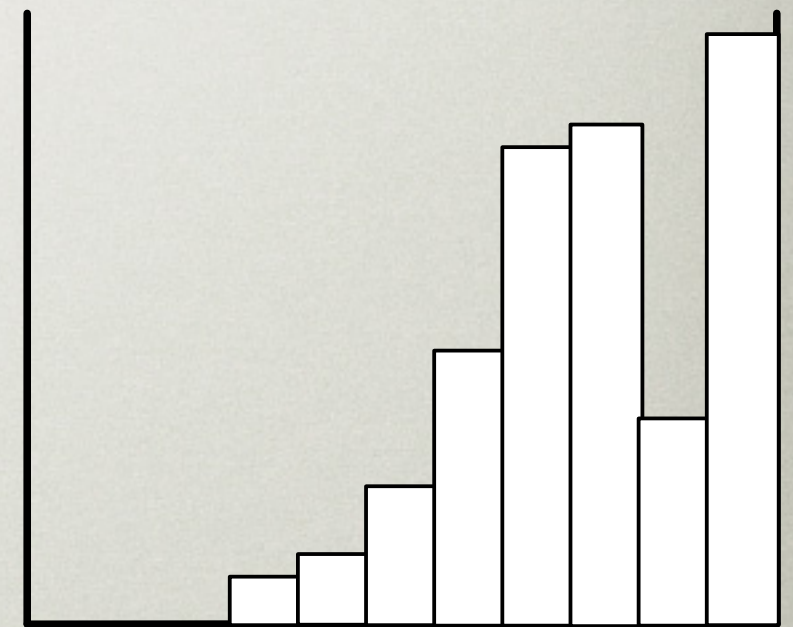
- Inspection of the GLH can show up faulty digitizer settings



(a) Insufficient signal



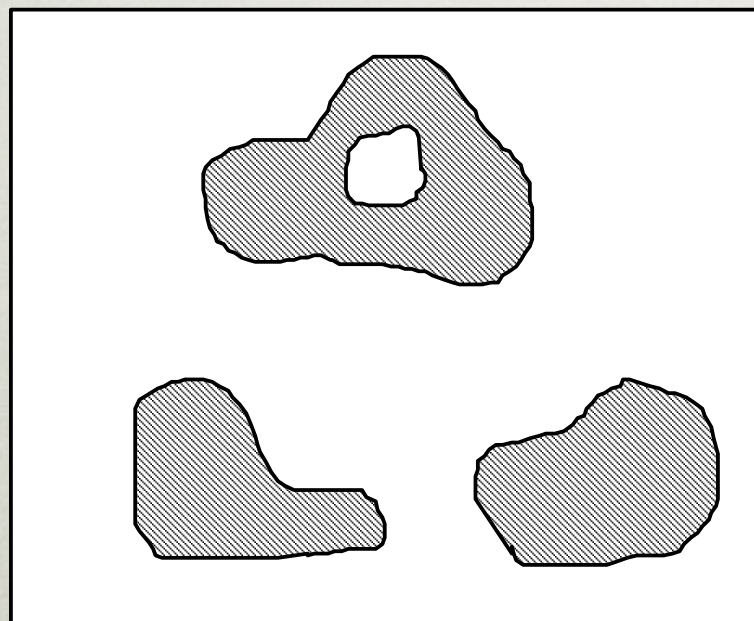
(b) Correct adjustment



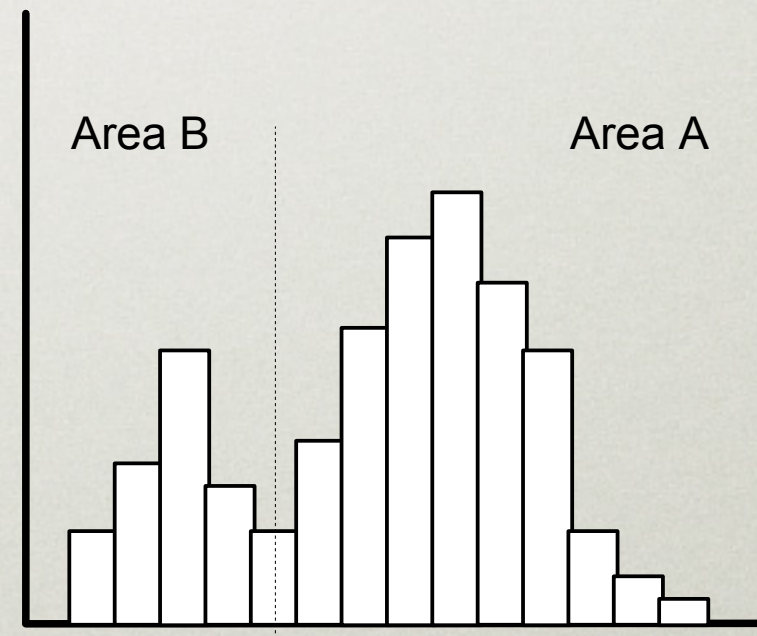
(c) Saturation

IMAGE SEGMENTATION

- The GLH can often be used to distinguish simple objects from background and determine their area.

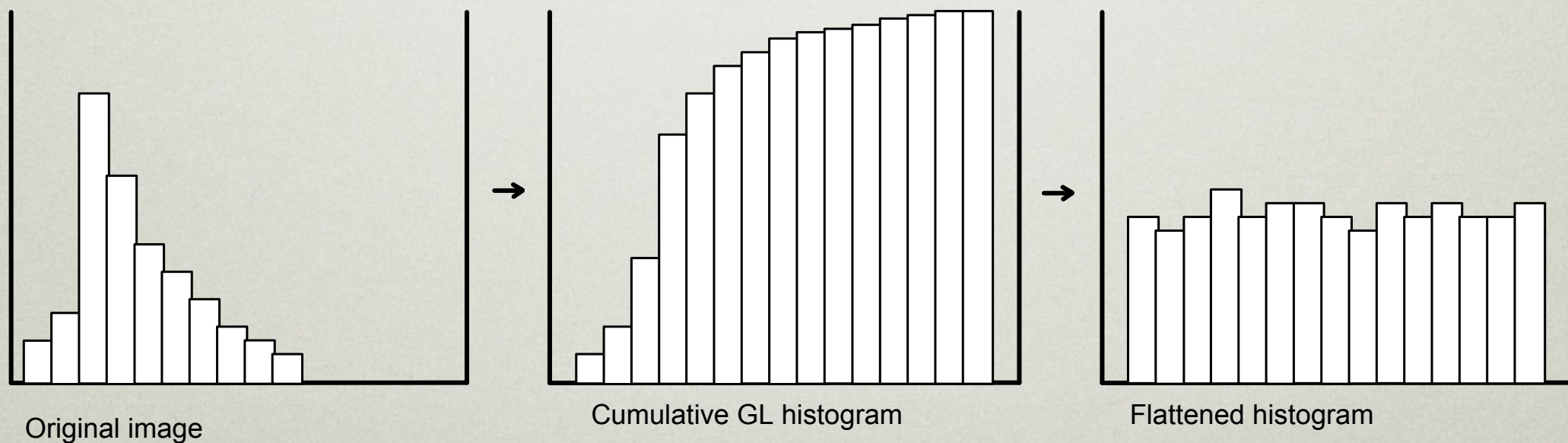


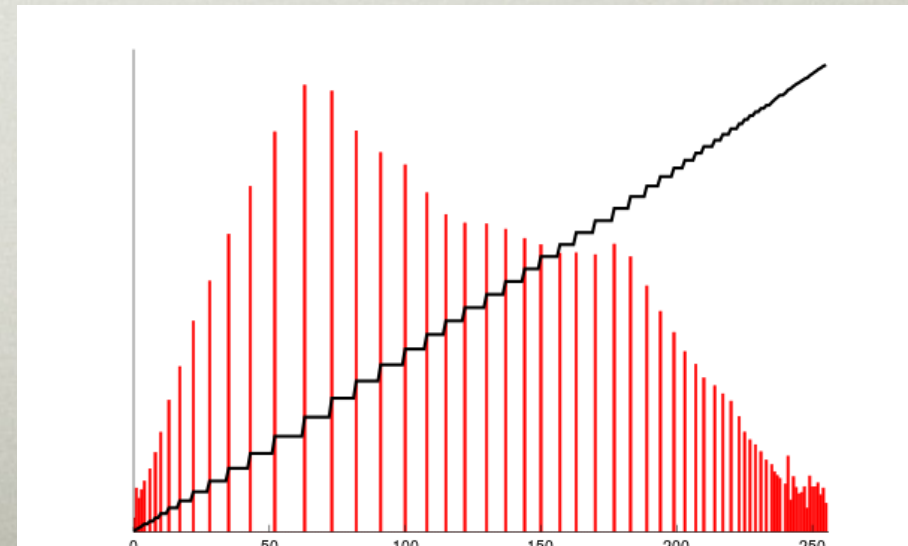
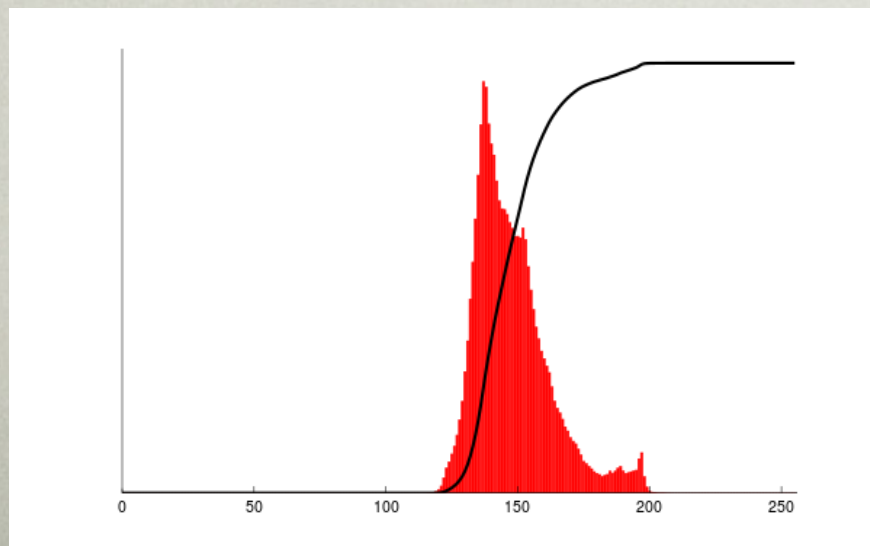
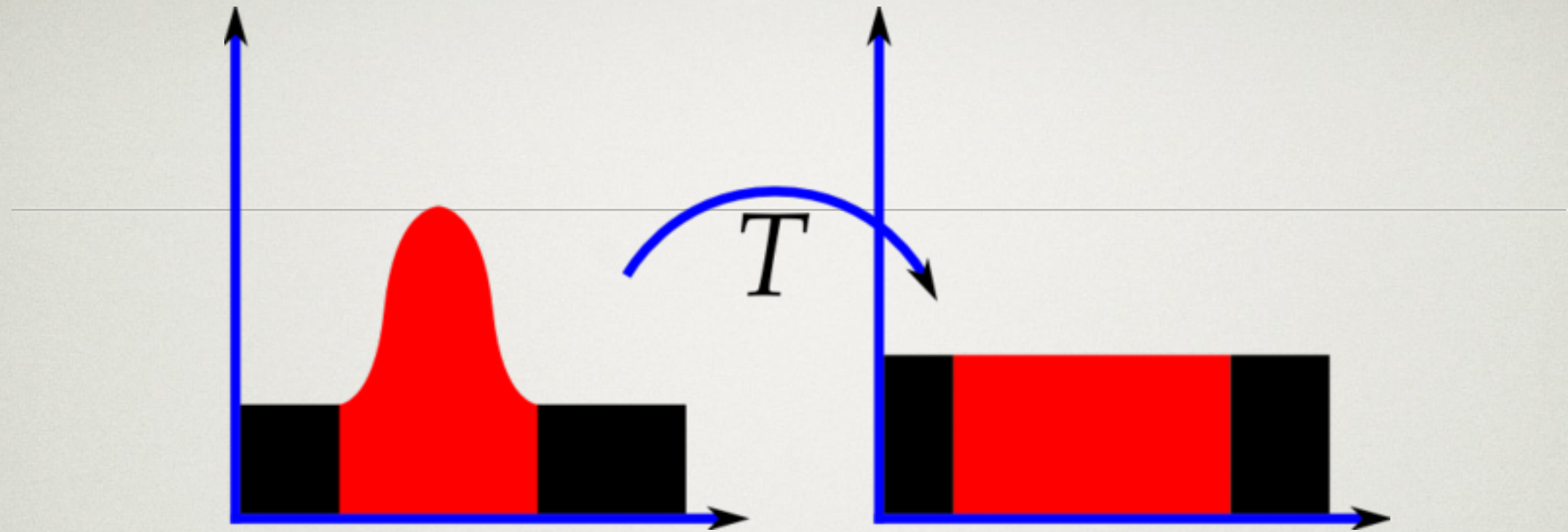
Dark objects, bright background



GREY LEVEL HISTOGRAM EQUALIZATION

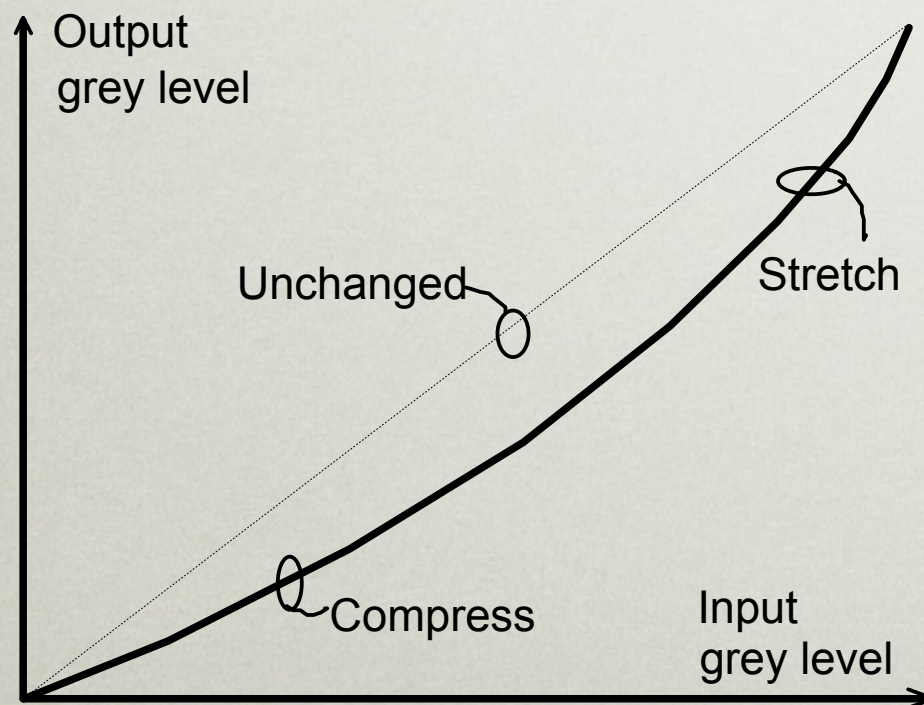
- In GLH equalization, a non-linear grey scale transformation redistributes the grey levels, producing an image with a flattened histogram.
- This can result in a striking contrast improvement.





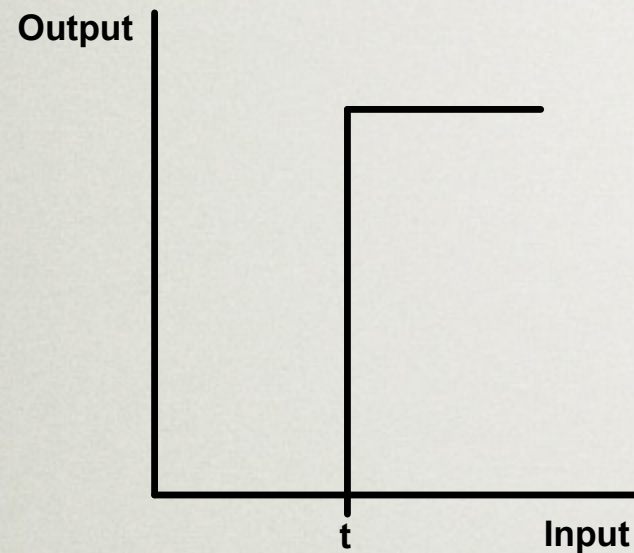
POINT OPERATIONS

- Point operations affect the way images occupy greyscale.
- A point operation transforms an input image, producing an output image in which each pixel grey level is related in a systematic way to that of the corresponding input pixel.

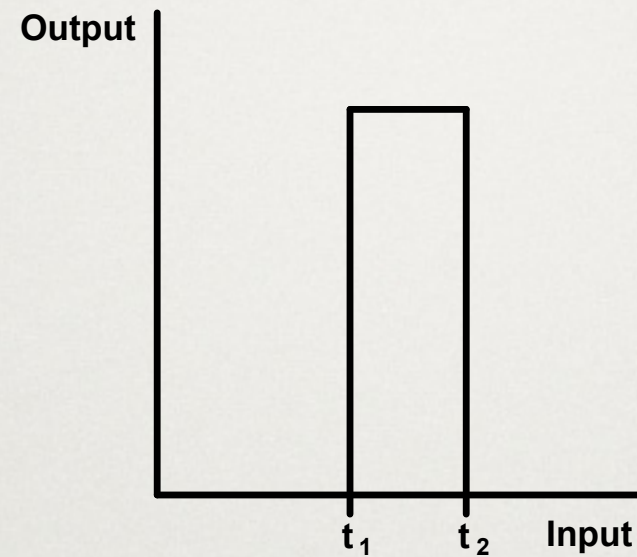


A point operation will never alter the spatial relationships within an image.

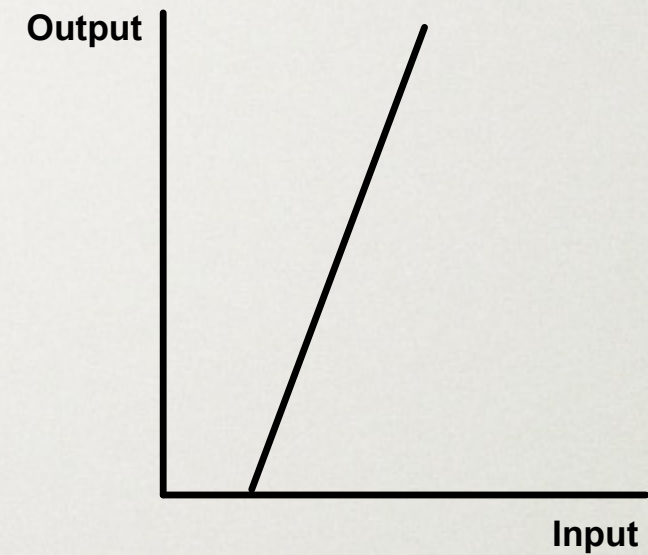
EXAMPLES OF POINT OPERATIONS



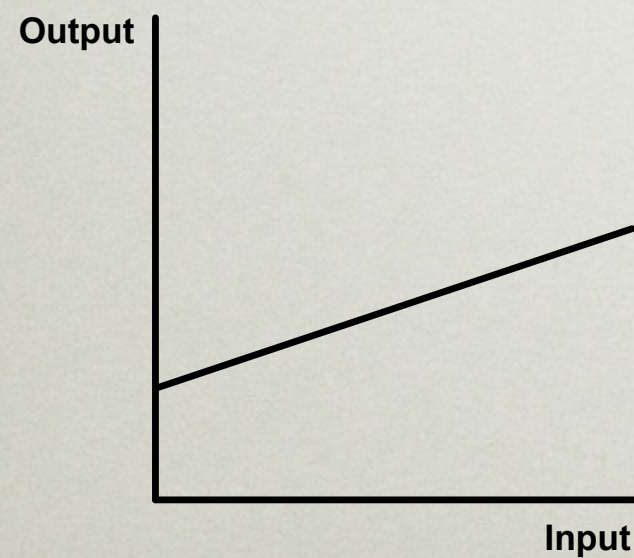
(a) Threshold



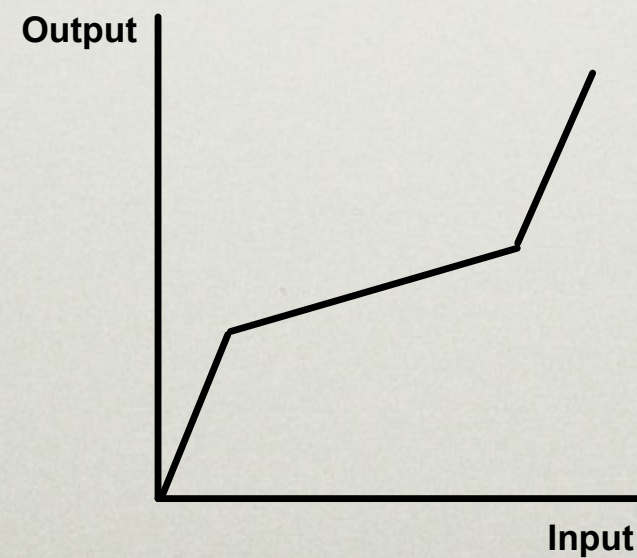
(b) Window Threshold



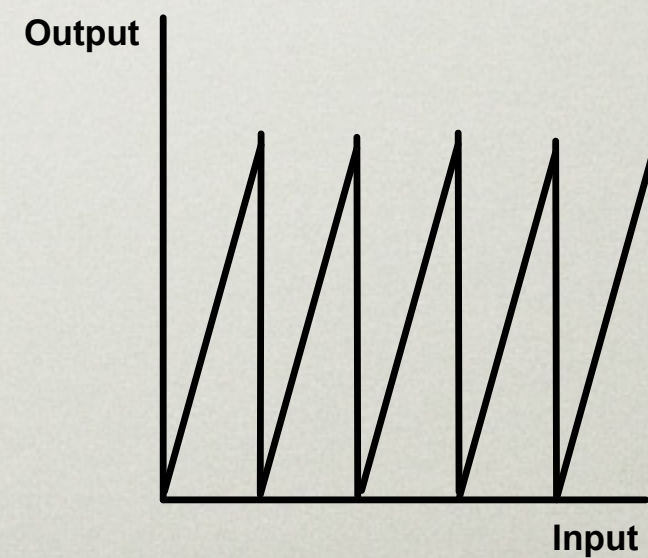
(c) Contrast Stretch



(d) Contrast Compression



(e) Combination



(f) Contouring

ALGEBRAIC OPERATIONS

- A point form of operation (with >1 input image)
- Grey level of each output pixel depends only on grey levels of corresponding pixels in input images

- Four major operations:

– Addition:

$$C = A + B$$

– Multiplication:

$$C = A \times B$$

– Subtraction

$$C = A - B$$

– Division:

$$C = A \div B$$

Other operations can be defined that involve more than 2 input images, or using (for example) boolean or logic operators.

APPLICATIONS OF ALGEBRAIC OPERATIONS

- Addition

- Ensemble averaging to reduce noise
- Superimposing one image upon another

- Subtraction

- Removal of unwanted additive interference (background suppression)
- Motion detection

APPLICATIONS (CONTINUED)

- Multiplication

- Removal of unwanted multiplicative interference (background suppression)
- Masking prior to combination by addition
- Windowing prior to Fourier transformation

- Division

- Background suppression (as multiplication)
- Special imaging signals (multi-spectral work)

NOISE REDUCTION

- An important noise reduction technique is frame averaging.
- A number of frames are averaged. The random noise is averaged out, resulting in a much improved image of the sample. Frame averaging may be written as:

$$y = \frac{1}{N} \sum_{n=1}^N x_n$$

where N is the number of images averaged, x are the images to be averaged and y is the averaged image.

LOCAL OPERATIONS

- In a local operation, the value of a pixel in the output image is a function of the corresponding pixel in the input image and its neighboring pixels. Local operations may be used for:-
 - image smoothing
 - noise cleaning
 - edge enhancement
 - boundary detection
 - assessment of texture

LOCAL OPERATIONS FOR IMAGE SMOOTHING

- Image averaging can be described as follows:-

1/4

0	1	0
1	0	1
0	1	0

$h(x, y)$

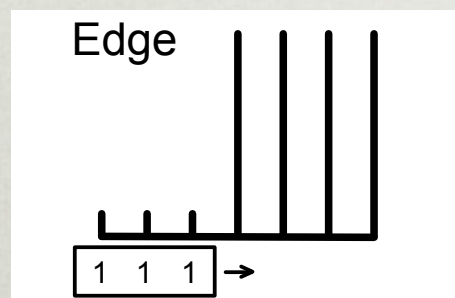
The mask shows graphically the disposition and weights of the pixels involved in the operation.

$$\text{Total weight} = \sum_n h(x, y) = 4$$

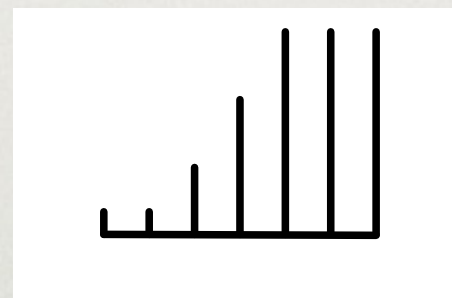
Image averaging is an example of low-pass filtering.

LOW PASS AND MEDIAN FILTERS

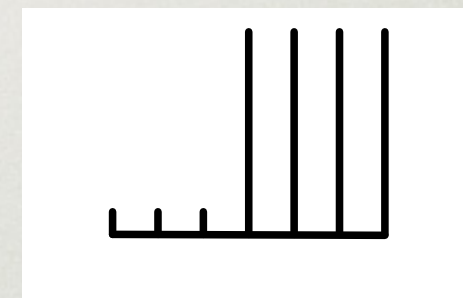
- The low-pass filter can provide image smoothing and noise reduction, but subdues and blurs sharp edges.
- Median filters can provide noise filtering without blurring.



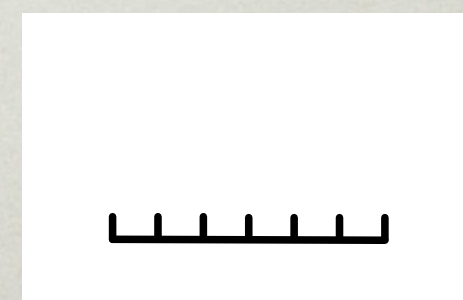
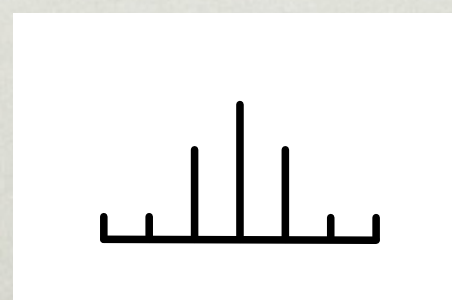
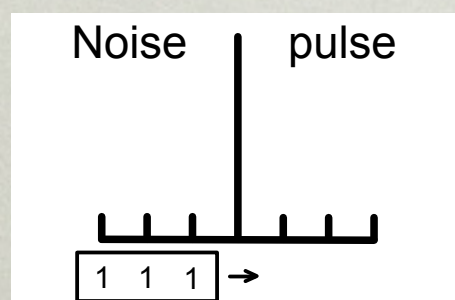
Original image



Averaging filter



Median filter



HIGH PASS FILTERS

- Subtracting contributions from neighboring pixels resembles differentiation, and can emphasize or sharpen variations in contrast. This technique is known as High Pass Filtering.

-1
1

h1

1	-1
---	----

h2

The simplest high-pass filter simulates the mathematical gradient operator:

$$G = \begin{bmatrix} df / dy \\ df / dx \end{bmatrix}$$

h1 gives the vertical, and h2 the horizontal component. The two parts are then summed (ignoring sign) to give the result.

FURTHER EXAMPLES OF FILTERS

- These masks contain 9 elements organised as 3 x 3. Calculation of one output pixel requires 9 multiplications & 9 additions

1/4

1	1	1
1	1	1
1	1	1

(a) Averaging

-1	0	1
-2	0	2
-1	0	1

(b) Sobel

0	1	0
1	-4	1
0	1	0

(c) Laplacian

-1	-1	-1
-1	9	-1
-1	-1	-1

(d) High Pass

FREQUENCY METHODS

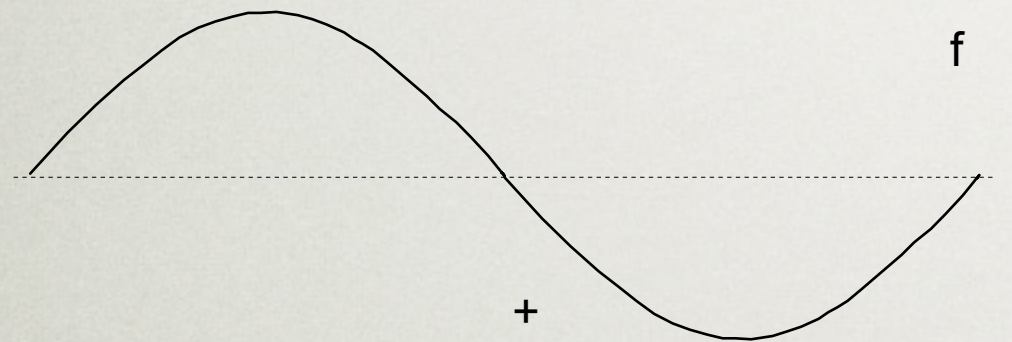
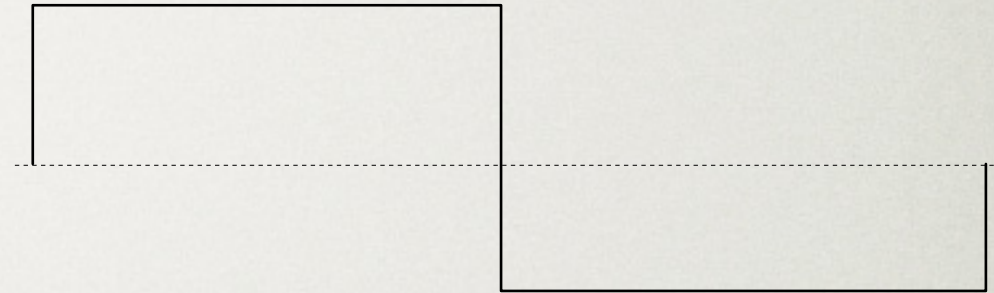
- Introduction to Frequency Domain
- The Fourier Transform
- Fourier filtering
- Example of Fourier filtering

FREQUENCY DOMAIN

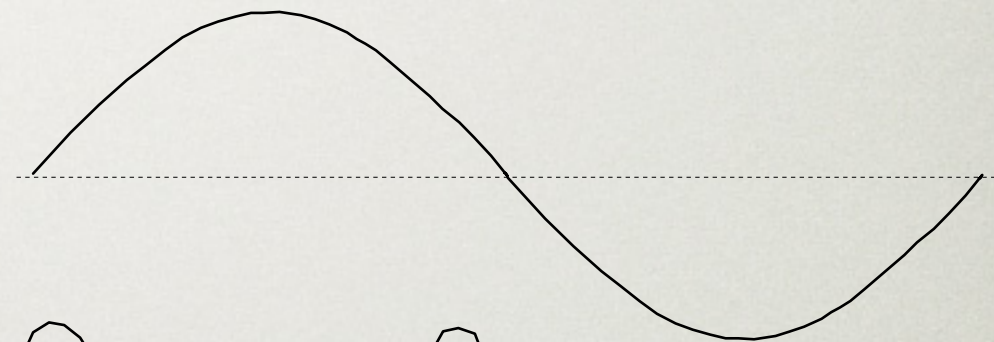
- Frequency refers to the rate of repetition of some periodic event. In imaging, Spatial Frequency refers to the variations of image brightness with position in space.
- A varying signal can be transformed into a series of simple periodic variations. The Fourier Transform is a well known example and decomposes the signal into a set of sine waves of different characteristics (frequency and phase).

THE FOURIER TRANSFORM

Original square wave



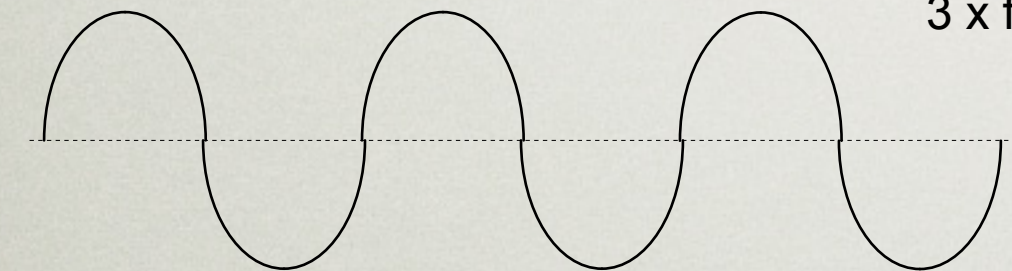
=



+

$3 \times f$

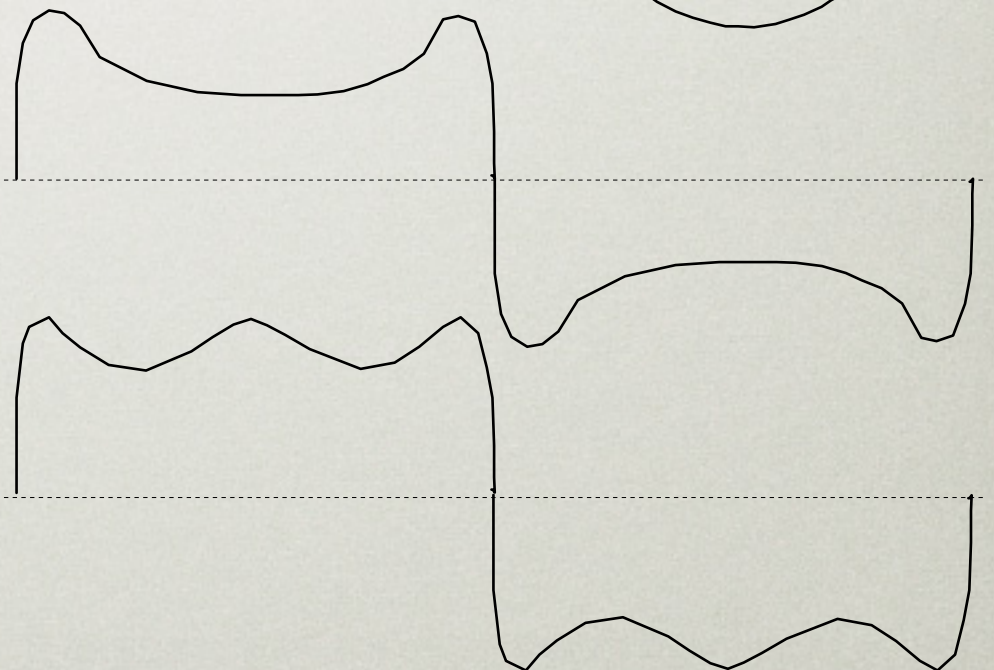
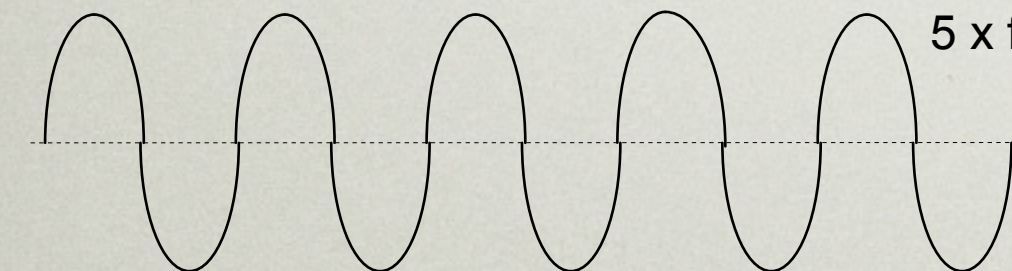
=



+

$5 \times f$

=



FOURIER FILTERING

- Discrete Fourier Transform (DFT) method.
- The DFT also allows spectral data (i.e. a transformed image) to be inverse transformed, producing an image once again.
- If we compute the DFT of an image, then immediately inverse transform the result, we expect to regain the same image.
- If we multiply each element of the DFT of an image by a suitably chosen weighting function we can accentuate certain frequency components and attenuate others. The corresponding changes in the spatial form can be seen after the inverse DFT has been computed.
- The selective enhancement/suppression of frequency components like this is known as Fourier Filtering.

USES OF FOURIER FILTERING

- Convolution with large masks (Convolution Theorem)
- Compensate for known image defects (restoration)
- Reduction of image noise
- Suppression of 'hum' or other periodic interference
- Reconstruction of 3D data from 2D sections
- Many others . . .

TRANSFORMS & IMAGE COMPRESSION

- Image transforms convert the spatial information of the image into a different form e.g. fast Fourier transform (F.F.T.) and discrete cosine transform (D.C.T.). A value in the output image is dependent on all pixels of the input image. The calculation of transforms is very computationally intensive.
- Image compression techniques reduce the amount of data required to store a particular image. Many of the image compression algorithms rely on the fact the eye is unable to perceive small changes in an image.

OTHER APPLICATIONS

- Image restoration (compensate instrumental aberrations)
- Lattice averaging & structure determination (esp. TEM)
- Automatic focussing & astigmatism correction
- Analysis of diffraction (and other related) patterns
- 3D measurements, visualization & reconstruction
- Analysis of sections (stereology)
- Image data compression, transmission & access
- Desktop publishing & multimedia

IMAGE ANALYSIS

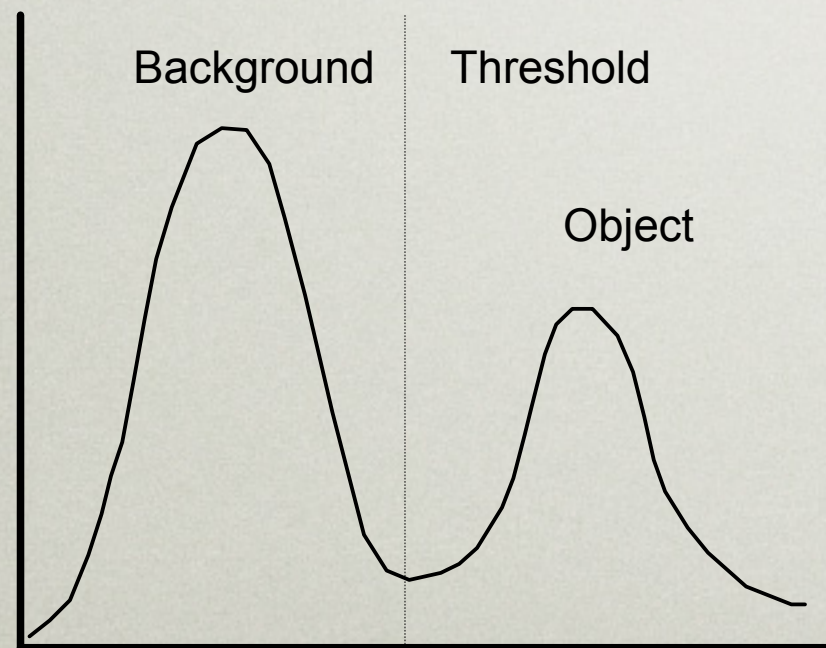
- Segmentation
 - Thresholding
 - Edge detection
- Representation of objects
- Morphological operations

SEGMENTATION

- Point-dependent methods
 - Thresholding and semi-thresholding
 - Adaptive thresholding
- Neighbourhood-dependent
 - Edge enhancement & edge detectors
 - Boundary tracking
 - Template matching

POINT-DEPENDENT METHODS

- Operate by locating groups of pixels with similar properties.
- Thresholding
- Assign a threshold grey level which discriminates between objects and background. This is straightforward if the image has a bimodal grey-level histogram.

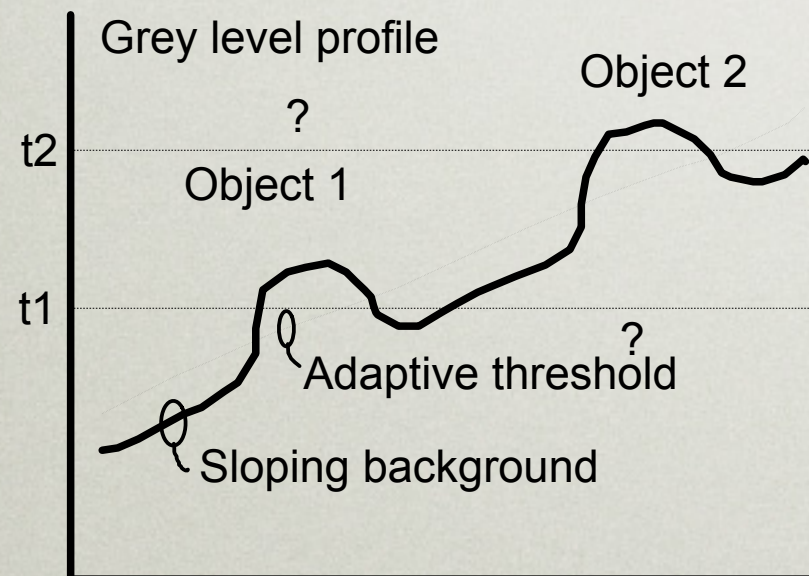


$$f_t = \begin{cases} 255 & f \geq t \\ 0 & f < t \end{cases} \quad (\text{thresholding})$$

$$f_t = \begin{cases} f & f \geq t \\ 0 & f < t \end{cases} \quad (\text{semi-thresholding})$$

ADAPTIVE THRESHOLDING

- In practice the GLH is rarely bimodal, owing to:-
 - Random noise - use LP / median or temporal filtering
 - Varying illumination
 - Complex images - objects of different sizes / properties



Background correction (subtract or divide) may be applied if an image of the background alone is available. Otherwise an adaptive strategy can be used.

NEIGHBOURHOOD-DEPENDENT OPERATIONS

- Edge detectors
 - Highlight region boundaries.
- Template matching
 - Locate groups of pixels in a particular group or configuration (pattern matching)
- Boundary tracking
 - Locate all pixels lying on an object boundary

EDGE DETECTORS

- Most edge enhancement techniques based on HP filters can be used to highlight region boundaries - e.g. Gradient, Laplacian. Several masks have been devised specifically for this purpose, e.g. Roberts and Sobel operators.
- Must consider directional characteristics of mask
- Effects of noise may be amplified
- Certain edges (e.g. texture edge) not affected

TEMPLATE MATCHING

- A template is an array of numbers used to detect the presence of a particular configuration of pixels. They are applied to images in the same way as convolution masks.

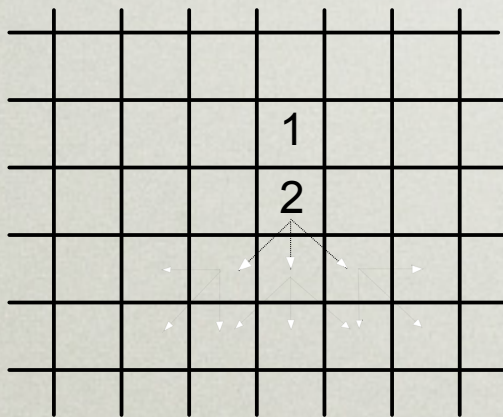
-1	-1	-1
-1	8	-1
-1	-1	-1

This 3x3 template will identify isolated objects consisting of a single pixel differing in grey-level from the background.

Other templates can be devised to identify lines or edges in chosen orientations.

BOUNDARY TRACKING

- Boundary tracking can be applied to any image containing only boundary information. Once a single boundary point is found, the operation seeks to find all other pixels on that boundary. One approach is shown:-



- Find first boundary pixel (1);
- Search 8 neighbours to find (2);
- Search in same direction (allow deviation of 1 pixel either side);
- Repeat step 3 till end of boundary.

CONNECTIVITY AND CONNECTED OBJECTS

- Rules are needed to decide to which object a pixel belongs.
- Some situations easily handled, others less straightforward.
- It is customary to assume either:
 - 4-connectivity
 - a pixel is regarded as connected to its four nearest neighbours
 - 8-connectivity.
 - a pixel is regarded as connected to all eight nearest neighbours

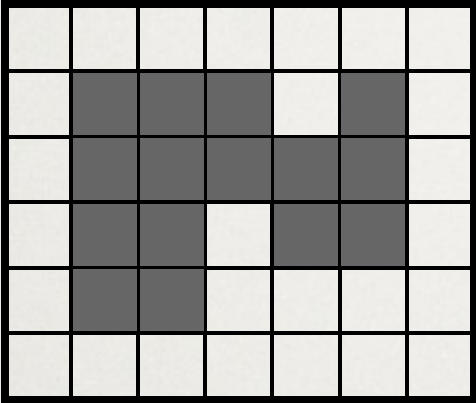
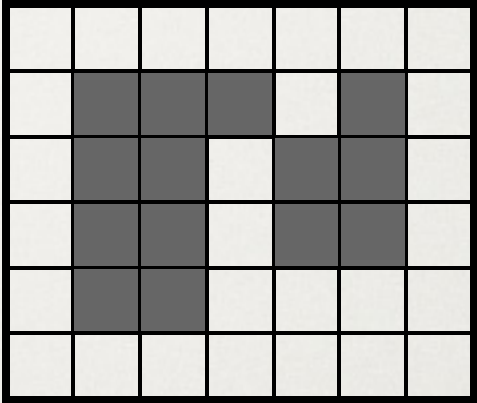
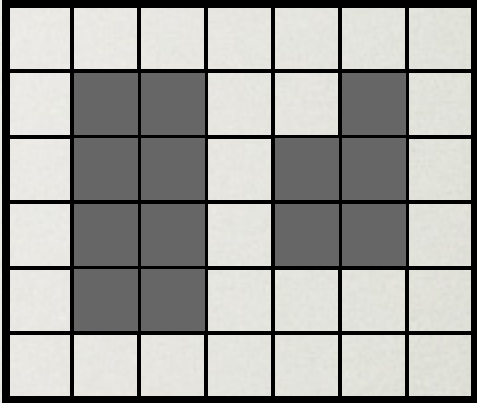
	1	
2	X	0
	3	

4-connected pixels

3	2	1
4	X	0
5	6	7

8-connected pixels

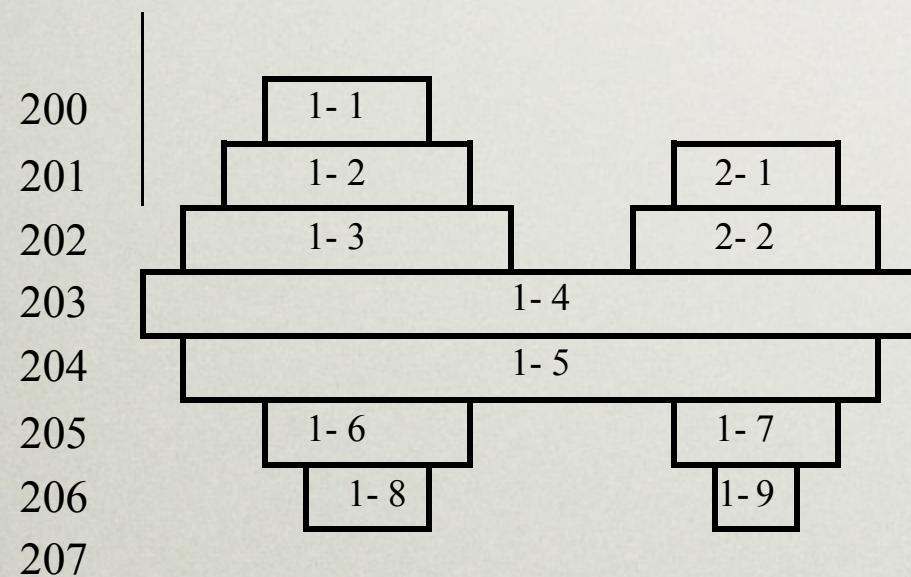
CONNECTED COMPONENTS

Object - Shaded							
Background - Blank							
Connectivity		A. Objects found		B. Objects found		C. Objects found	
4		1		2		2	
8		1		1		2	

- Results of analysis under 4- or 8- connectivity
- A hidden paradox affects object and background pixels

LINE SEGMENT ENCODING

- Objects are represented as collections of chords
- A line-by-line technique
- Requires access to just two lines at a time



1-1 is the first segment found.
 1-2 & 2-1 appear to be separate objects.
 1-4, 1-5 underlie 1-3, 2-2;
 Hence, 1- & 2- are the same

1-6 to 1-9 belong to same object.

Obj label Segments Segment labels...

19	11	1-1	1-2	..
----	----	-----	-----	----

..	202	311	8	..
Row		Col	Pixels	

REPRESENTATION OF OBJECTS

- **Object membership map (OMM)**
- An image the same size as the original image
- Each pixel encodes the corresponding object number, e.g. all pixels of object 9 are encoded as value 9
- Zero represents background pixels

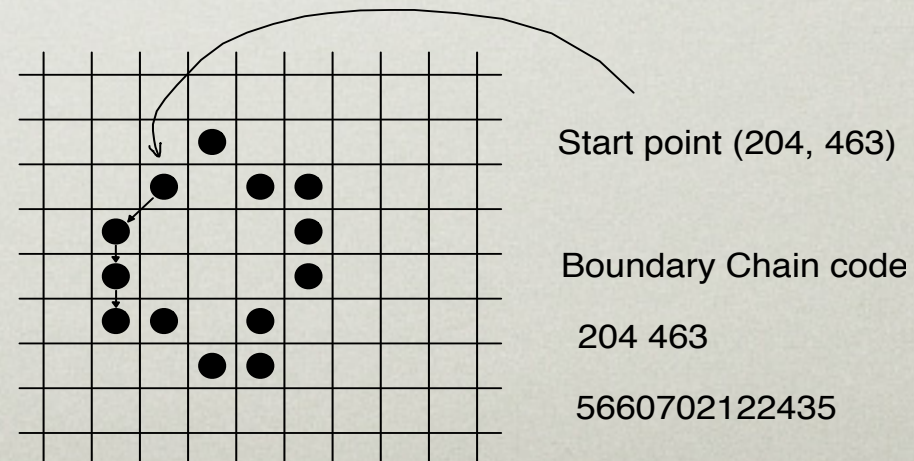
Example OMM

[illegible]

REPRESENTATION OF OBJECTS

- A compact format for storing object information about an object
- Defines only the position of the object boundary
- Takes advantage of connected nature of boundaries.
- Economical representation; 3 bits/boundary point
- Yields some feature information directly
 - Choose a starting point on the boundary (arbitrary)
 - One or more nearest neighbours must also be a boundary point
 - Record the direction codes that specify the path around the boundary

3	2	1
4	■	0
5	6	7



SIZE MEASUREMENTS

- **Area**
- A simple, convenient measurement, can be determined during extraction.
- The object pixel count, multiplied by the area of a single pixel.
 - Determined directly from the segment-encoded representation
 - Additional computation needed for boundary chain code.

```
•      a = 0;                                // Initialise area to 0
•      x = n; y = n;                          // Arbitrary start coordinates
•      for (i=0; i<n; i++)
•      switch (c[i])                          // Inspect each element
•      {                                     // 0246 are parallel to the axes
•          case 0:a -= y;  x++; break;
•          case 2:      y++; break;
•          case 4:a += y;  x--; break;
•          case 6:      y--; break;
•      }
```

LENGTH AND WIDTH

Straightforwardly computed during encoding or tracking.

- Record coordinates:
 - minimum x
 - maximum x
 - minimum y
 - maximum y
- Take differences to give:
 - horizontal extent
 - vertical extent
 - minimum boundary rectangle.

NUMBER OF HOLES

Hole count may be of great value in classification.

A fundamental relationship exists between:-

- the number of connected components C (i.e. objects)
- the number of holes H in a figure
- and the *Euler number*:-

$$E = C - H$$

A number of approaches exist for determining H .

- Count special motifs (known as *bit quads*) in objects.

These can give information about:-

- Area
- Perimeter
- Euler number

DERIVED FEATURES

For example, shape features

Rectangularity

Ratio of object area A to area A_E of minimum enclosing rectangle

- Expresses how efficiently the object fills the MER
- Value must be between 0 and 1.
- For circular objects it is
- Becomes small for curved, thin objects.

Aspect ratio

The width/length ratio of the minimum enclosing rectangle

- Can distinguish slim objects from square/circular objects

DERIVED FEATURES (CONT)

Circularity

- Assume a minimum value for circular shape
- High values tend to reflect complex boundaries.

One common measure is:

$$C = P^2 / A$$

(ratio of perimeter squared to area)

- takes a minimum value of 4π for a circular shape.
- Warning: value may vary with magnification

TEXTURE ANALYSIS

Repetitive structure cf. tiled floor, fabric

How can this be analyzed quantitatively?

One possible solution based on edge detectors:-

- determine orientation of gradient vector at each pixel
- quantize to (say) 1 degree intervals
- count the number of occurrences of each angle
- plot as a polar histogram:
 - radius vector $\propto$ number of occurrences
 - angle corresponds to gradient orientation
- Amorphous images give roughly circular plots
- Directional, patterned images may give elliptical plots

Summary